



8.5.2 Summary ¶

Given a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, its **gradient** is given by

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_0}(x) \\ \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_{n-1}}(x) \end{pmatrix}.$$

$\nabla f(x)$ equals the direction in which the function f increases most rapidly at the point x and $-\nabla f(x)$ equals the direction of steepest descent (the direction in which the function f decreases most rapidly at the point x).

In this summary, we will assume that $A \in \mathbb{R}^{n \times n}$ is symmetric positive definite (SPD) and

$$f(x) = \frac{1}{2}x^T A x - x^T b.$$

The gradient of this function equals

$$\nabla f(x) = Ax - b$$

and $\hat{x}$ minimizes the function if and only if

$$A\hat{x} = b.$$

If $x^{(k)}$ is an approximation to $\hat{x}$ then $r^{(k)} = b - Ax^{(k)}$ equals the corresponding residual. Notice that $r^{(k)} = -\nabla f(x^{(k)})$ and hence $r^{(k)}$ is the direction of steepest descent.

A prototypical descent method is given by

```
Given :  $A, b, x^{(0)}$   
 $r^{(0)} := b - Ax^{(0)}$   
 $k := 0$   
while  $r^{(k)} \neq 0$   
   $p^{(k)} :=$  next direction  
   $x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$  for some scalar  $\alpha_k$   
   $r^{(k+1)} := b - Ax^{(k+1)}$   
   $k := k + 1$   
endwhile
```

Here $p^{(k)}$ is the "current" search direction and in each iteration we create the next approximation to $\hat{x}$, $x^{(k+1)}$, along the line $x^{(k)} + \alpha p^{(k)}$.

If $x^{(k+1)}$ minimizes along that line, the method is an exact descent method and

$$\alpha_k = \frac{p^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}$$

so that a prototypical exact descent method is given by

```
Given :  $A, b, x^{(0)}$   
 $r^{(0)} := b - Ax^{(0)}$   
 $k := 0$   
while  $r^{(k)} \neq 0$   
   $p^{(k)} :=$  next direction  
   $\alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}$   
   $x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$   
   $r^{(k+1)} := b - Ax^{(k+1)}$   
   $k := k + 1$   
endwhile
```

Once α_k is determined,

$$r^{(k+1)} = r^{(k)} - \alpha_k A p^{(k)}.$$

which saves a matrix-vector multiplication when incorporated into the prototypical exact descent method:

```
Given :  $A, b, x^{(0)}$   
 $r^{(0)} := b - Ax^{(0)}$   
 $k := 0$   
while  $r^{(k)} \neq 0$   
   $p^{(k)} :=$  next direction  
   $q^{(k)} := A p^{(k)}$   
   $\alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} q^{(k)}}$   
   $x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$   
   $r^{(k+1)} := r^{(k)} - \alpha_k q^{(k)}$   
   $k := k + 1$   
endwhile
```

The steepest descent algorithm chooses $p^{(k)} = -\nabla f(x^{(k)}) = b - Ax^{(k)} = r^{(k)}$:

```

Given :  $A, b, x^{(0)}$ 
 $r^{(0)} := b - Ax^{(0)}$ 
 $k := 0$ 
while  $r^{(k)} \neq 0$ 
   $p^{(k)} := r^{(k)}$ 
   $q^{(k)} := Ap^{(k)}$ 
   $\alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} q^{(k)}}$ 
   $x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$ 
   $r^{(k+1)} := r^{(k)} - \alpha_k q^{(k)}$ 
   $k := k + 1$ 
endwhile

```

Convergence can be greatly accelerated by incorporating a preconditioner, M , where, ideally, $M \approx A$ is SPD and solving $Mz = y$ is easy (cheap).

```

Given :  $A, b, x^{(0)}, M$ 
 $r^{(0)} := b - Ax^{(0)}$ 
 $k := 0$ 
while  $r^{(k)} \neq 0$ 
   $p^{(k)} := M^{-1} r^{(k)}$ 
   $q^{(k)} := Ap^{(k)}$ 
   $\alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} q^{(k)}}$ 
   $x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$ 
   $r^{(k+1)} := r^{(k)} - \alpha_k q^{(k)}$ 
   $k := k + 1$ 
endwhile

```

Definition 8.5.2.1. A-conjugate directions. Let A be SPD. A sequence $p^{(0)}, \dots, p^{(k-1)} \in \mathbb{R}^n$ such that $p^{(j)T} A p^{(i)} = 0$ if and only if $j \neq i$ is said to be A-conjugate.

The columns of $P \in \mathbb{R}^{n \times k}$ are A-conjugate if and only if $P^T A P = D$ where D is diagonal and has positive values on its diagonal.

A-conjugate vectors are linearly independent.

A descent method that chooses the search directions to be A-conjugate will find the solution of $Ax = b$, where $A \in \mathbb{R}^{n \times n}$ is SPD, in at most n iterations:

```

Given :  $A, b$ 
 $x^{(0)} := 0$ 
 $r^{(0)} = b$ 
 $k := 0$ 
while  $r^{(k)} \neq 0$ 
  Choose  $p^{(k)}$  such that  $p^{(k)T} A p^{(k-1)} = 0$  and  $p^{(k)T} r^{(k)} \neq 0$ 
   $\alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}$ 
   $x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$ 
   $r^{(k+1)} := r^{(k)} - \alpha_k A p^{(k)}$ 
   $k := k + 1$ 
endwhile

```

The Conjugate Gradient Method chooses the search direction to equal the vector $p^{(k)}$ that is A-conjugate to all previous search directions and is closest to the direction of steepest descent:

```

Given :  $A, b$ 
 $x^{(0)} := 0$ 
 $r^{(0)} := b$ 
 $k := 0$ 
while  $r^{(k)} \neq 0$ 
  if  $k = 0$ 
     $p^{(k)} = r^{(0)}$ 
  else
     $p^{(k)}$  minimizes  $\min_{p \perp \text{Span}(A p^{(0)}, \dots, A p^{(k-1)})} \|r^{(k)} - p\|_2$ 
  endif
   $\alpha_k := \frac{p^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}$ 
   $x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$ 
   $r^{(k+1)} := r^{(k)} - \alpha_k A p^{(k)}$ 
endwhile

```

The various vectors that appear in the Conjugate Gradient Method have the following properties: If $P^{(p-1)} = (p^{(0)} \dots p^{(k-1)})$ then

- $P^{(k-1)T} r^{(k)} = 0$.
- $\text{Span}(p^{(0)}, \dots, p^{(k-1)}) = \text{Span}(r^{(0)}, \dots, r^{(k-1)}) = \text{Span}(b, Ab, \dots, A^{k-1}b)$.
- The residual vectors $r^{(k)}$ are mutually orthogonal.
- For $k \geq 1$

$$p^{(k)} = r^{(k)} - \gamma_k p^{(k-1)}$$

Definition 8.5.2.2. Krylov subspace. The subspace

$$\mathcal{K}_k(A, b) = \text{Span}(b, Ab, \dots, A^{k-1}b)$$

is known as the **order- k Krylov subspace**.

Alternative Conjugate Gradient Methods are given by

Given : A, b

$$x^{(0)} := \mathbf{0}$$
$$r^{(0)} := b$$
$$k := 0$$
while $r^{(k)} \neq 0$

if $k = 0$

$$p^{(k)} = r^{(0)}$$

else

$$\gamma_k := (\mathbf{r}^{(k)T} \mathbf{r}^{(k)}) / (\mathbf{r}^{(k-1)T} \mathbf{r}^{(k-1)})$$
$$p^{(k)} := r^{(k)} + \gamma_k p^{(k-1)}$$

endif

$$\alpha_k := \frac{r^{(k)T} r^{(k)}}{p^{(k)T} A p^{(k)}}$$
$$x^{(k+1)} := x^{(k)} + \alpha_k p^{(k)}$$
$$r^{(k+1)} := r^{(k)} - \alpha_k A p^{(k)}$$
$$k := k + 1$$

endwhile

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